

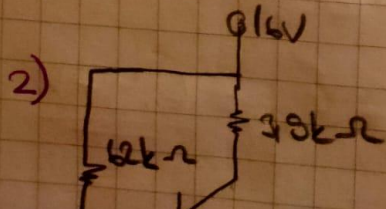
$$I_b, I_c, V_{CE}, V_E = ?$$

$$I_b = \frac{15 - 0.7}{200k} = \underline{\underline{71.5 \mu A}}$$

$$I_c = I_b \cdot \beta = 71.5 \mu A \cdot 90 = \underline{\underline{6.43 mA}}$$

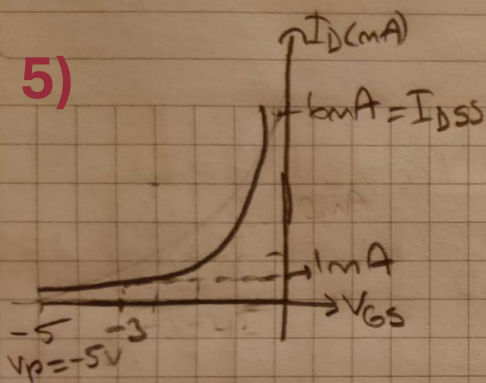
$$V_{CE} = 15 - \frac{1k \cdot 6.43 mA}{6.43V} = \underline{\underline{8.57V}}$$

$$\underline{\underline{V_E = 0}}$$



Yaklaşık analiz ile $V_B, V_E, I_E, V_{CE}, V_C$ bul

5. Soru Grafik Çözümü:



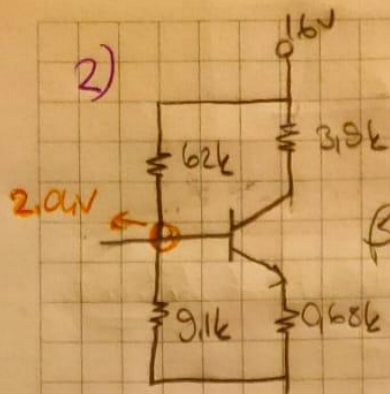
$$\underline{\underline{I_D = 1 mA}}$$

$$\underline{\underline{V_{GS} = -2V}}$$

$$V_{DS} = 20 - 1mA \cdot 6.8k\Omega$$

$$\underline{\underline{V_{DS} = 13.2V = V_D}}$$

$$\underline{\underline{V_S = 0}}$$



Yaklaşık analiz ile $V_B, V_E, I_E, V_{CE}, V_C$ bulunuz.

$$\beta = 80 \quad V_B = \frac{16V \cdot 9.1k}{62k + 9.1k} = \underline{2.04V}$$

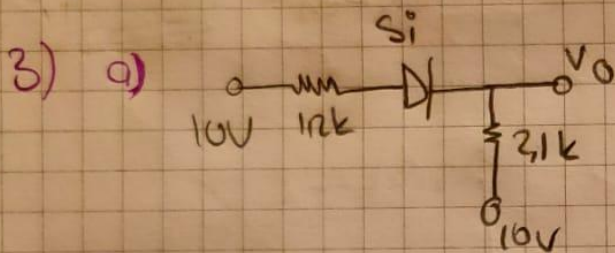
$$I_E = \frac{2.04 - 0.7}{0.68k} = \underline{1.97mA}$$

$$V_E = 1.97mA \cdot 0.68k = \underline{1.34V}$$

$$I_E \approx I_C$$

$$V_C = 16 - 3.9k \cdot I_C = \underline{8.3V}$$

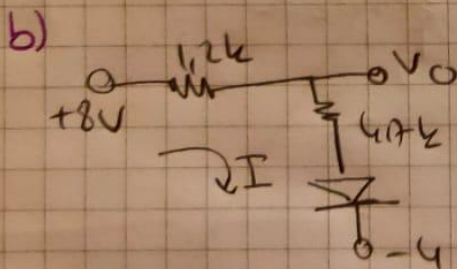
$$V_{CE} = V_C - V_E = \underline{6.97V}$$



I ve V_O 'yu bulunuz.

$$\underline{I = 0} \text{ iletimde değil}$$

$$\underline{V_O = 10V}$$



I ve V_O 'yu bulunuz.

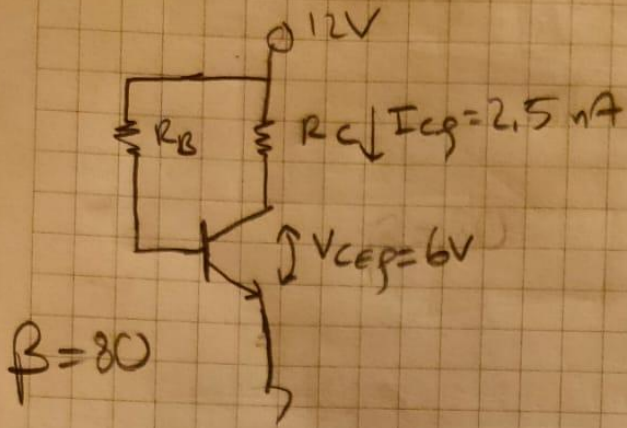
$$8 = 12k\Omega \cdot I + 4.7k\Omega \cdot I + 0.7 - 4$$

$$\underline{I = 1.81mA}$$

$$V_O = 8 - 12k\Omega \cdot 1.81mA$$

$$\underline{V_O = 5.7V}$$

4) Sabit öngerilimli yapıda (fixed bias) $V_{CC}=12V$, $\beta=80$
 $I_{CQ}=2.5mA$ $V_{CEQ}=6V$ ise R_C ve R_B 'yi bulunuz. (devreyi çizin)



$$V_{CE} = V_C = V_{CC} - I_{CQ} R_C$$

$$6V = 12 - 2.5mA \cdot R_C$$

$$\underline{\underline{R_C = 2.4k}}$$

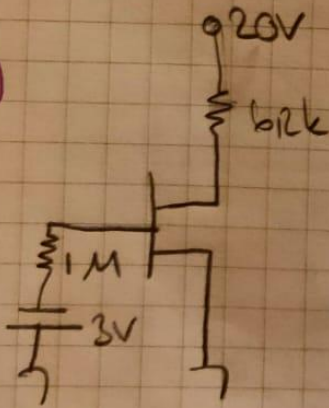
$$I_B = I_C / \beta$$

$$I_B = 2.5mA / 80$$

$$I_B = 312.5\mu A$$

$$R_B = \frac{12 - 0.7}{312.5\mu A} = \underline{\underline{361.6k}}$$

5)



$$I_{DSS} = 6mA$$

$$V_P = -5V$$

I_{DQ} , V_{GSQ} , V_D , V_S 'yi hesap ve grafiklerle bulunuz.

$$\underline{\underline{V_{GSQ} = -3V}}$$

$$I_{DQ} = I_{DSS} \left(1 - \frac{V_{GSQ}}{V_P}\right)^2$$

$$= 6mA \left(1 - \frac{-3}{-5}\right)^2$$

$$\underline{\underline{I_{DQ} = 360\mu A}}$$

$$20 - 6.2k \cdot 0.36mA - V_{DS} = 0$$

$$V_{DS} = 14.04V$$

$$\underline{\underline{V_S = 0V}} \Rightarrow \underline{\underline{V_D = 14.04V}}$$